LESSON 7.9

Reflect Points Across Axes

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

WHY THIS MATTERS Captain Vega discovers that her treasure map has a magic mirror trick! When she folds the map along the x-axis, some clues match up perfectly with clues on the other side. For example, a marker at (2, 4) has a matching marker at (2, -4) when reflected over the x-axis. Another pair reflects over the y-axis: (5, 3) and (-5, 3). How does reflection change the coordinates?

REMEMBER

Reflect Points Across Axes

Reflections Across Axes. Formula: Reflect over x-axis: (x, y) → (x, −y) | Reflect over y-axis: (x, y) → (−x, y) A reflection is a flip of a point over a line, like a mirror image. The axis you flip over decides which coordinate changes its sign.

1. Reflection across x-axis (horizontal line): y-coordinate changes sign, x stays same.
2. Reflection across y-axis (vertical line): x-coordinate changes sign, y stays same.
3. Both coordinates change sign when reflecting across both axes: (−x, −y)

Words: Reflection *(Reflexión)* | x-axis *(Eje x)* | y-axis *(Eje y)* | Symmetry *(Simetría)* | Integer *(Número entero)*

1 **Circle the letter of the best answer. Show how you know.**

In which quadrant is the point (−4, 5)?

(A) Quadrant I
(B) Quadrant II
(C) Quadrant III
(D) Quadrant IV

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which point lies ON the y-axis?

(A) (0, 3)
(B) (3, 0)
(C) (3, 3)
(D) (−3, 1)

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What are the coordinates of the origin?

(A) (0, 0)
(B) (0, 1)
(C) (1, 1)
(D) (1, 0)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A point has a negative x-coordinate and a negative y-coordinate. Which quadrant is it in?

(A) Quadrant I
(B) Quadrant II
(C) Quadrant III
(D) Quadrant IV

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A game designer creates a spaceship at (3, 2) and wants to place enemy ships as reflections to make the level symmetric. One enemy is reflected over the y-axis, and another is reflected over the x-axis. Where is the ship after reflecting (3, 2) over the y-axis?

(A) (2, 3)
(B) (−3, 2)
(C) (3, −2)
(D) (−3, −2)

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A game designer creates a spaceship at (3, 2) and wants to place enemy ships as reflections to make the level symmetric. One enemy is reflected over the y-axis, and another is reflected over the x-axis. Where is the ship after reflecting (3, 2) over the x-axis?

(A) (2, 3)
(B) (3, −2)
(C) (−3, 2)
(D) (−3, −2)

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

SENTENCE STARTER When I reflect over the x-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same. When I reflect over the y-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS Captain Vega discovers that her treasure map has a magic mirror trick! When she folds the map along the x-axis, some clues match up perfectly with clues on the other side. For example, a marker at (2, 4) has a matching marker at (2, -4) when reflected over the x-axis. Another pair reflects over the y-axis: (5, 3) and (-5, 3). How does reflection change the coordinates?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each map marker and fold line to its mirror image.

_____ (1, 7) folded over x-axis	A (1, -7)
_____ (-8, 2) folded over y-axis	B (8, 2)
_____ (9, -3) folded over x-axis	C (9, 3)
_____ (-1, -6) folded over y-axis	D (1, -6)

2 **Write the name of the correct group on each line.**

Sort each fold by which coordinate rule it follows on Vega's map.

Groups: **Over x-axis: keep x, flip y (x, -y)** **Over y-axis: flip x, keep y (-x, y)**

_____ (4, 9) fold over x-axis

_____ (-2, 5) fold over x-axis

_____ (6, -1) fold over x-axis

_____ (3, 8) fold over y-axis

_____ (-7, 4) fold over y-axis

_____ (2, -9) fold over y-axis

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A game designer creates a spaceship at (3, 2) and wants to place enemy ships as reflections to make the level symmetric. One enemy is reflected over the y-axis, and another is reflected over the x-axis. Reflecting over the y-axis changes which coordinate?

(A) Only the y-coordinate

(B) Only the x-coordinate

(C) Both coordinates

(D) Neither coordinate

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Captain Vega folds the map along the x-axis. A marker at (7, 2) lands on its mirror image. What are the new coordinates?

(A) (-7, 2)

(B) (-7, -2)

(C) (2, 7)

(D) (7, -2)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A buried chest is marked at (-5, 3). Vega folds the map along the y-axis to find its mirror twin. Where is the twin?

(A) (-5, -3)

(B) (3, -5)

(C) (5, 3)

(D) (5, -3)

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Vega reflects a palm-tree marker at (8, -6) across the x-axis. Which coordinate changes its sign?

(A) The x-coordinate

(B) The y-coordinate

(C) Both coordinates

(D) Neither coordinate

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

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